Heterogeneous Scientific Foundation Model Collaboration

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Research Questions

- How can domain-specific foundation models (e.g., time series, tabular data) collaborate within language-centric agentic systems without requiring full translation to natural language?
- Can heterogeneous foundation models achieve better performance than language-only agents on scientific tasks while reducing computational overhead?
- How should multi-agent systems dynamically orchestrate between language agents and specialized foundation model agents for complex scientific tasks?

Core Idea

Current agentic AI systems rely on language as the universal interface, but this creates a bottleneck for scientific tasks involving specialized data modalities like time series or molecular structures. The key insight is to create a 'neural bond' (inspired by Avatar's Tsaheylu) between LLMs and domain-specific foundation models through bidirectional interfaces: a query compiler that translates LLM reasoning into structured FM invocations, and a response adapter that converts FM outputs back into language-compatible representations. This allows specialized models to participate directly in reasoning without forcing everything through language tokenization, enabling modality-native collaboration while preserving the LLM's planning capabilities.

Key Formula

Domain Advantage Assumption

$$E_{\tau=(q,x,y^*,\ell)\sim T}[\ell_k(F_k(x_k),y^*)] < \inf_{A_{LLM}} E_{\tau=(q,x,y^*,\ell)\sim T}[\ell_k(A_{LLM}(\text{serialize}(x_k)),y^*)]$$

Formalizes that foundation model F_k achieves strictly better performance than any language model on domain-specific component x_k. This assumption justifies why specialized models should be preferred over language-based serialization for domain-specific data.

EywaAgent Risk Improvement

$$\inf_{f\in F_{Eywa}} E_{\tau\sim T}[\ell(f(x),y^*)] < \inf_{f\in F_{LLM}} E_{\tau\sim T}[\ell(f(x),y^*)]$$

Shows that EywaAgent function class F_Eywa achieves strictly lower optimal expected task loss than language-only function class F_LLM. Proves theoretical advantage of the hybrid approach over pure language agents.

Key Figure

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Code: https://github.com/Violet24K/Eywa [Project Page]

Abstract: Agentic large language model systems have demonstrated strong capabilities. However, their reliance on language as the universal interface fundamentally limits their applicability to many real-world problems, especially in scientific domains where domain-specific foundation models have been developed to address specialized tasks beyond natural language. In this work, we introduce *Eywa*, a heterogeneous agentic framework designed to extend language-centric systems to a broader class of scientific foundation models. The key idea of Eywa is to augment domain-specific foundation models with a language-model-based reasoning interface, enabling language models to guide inference over non-linguistic data modalities. This design allows predictive foundation models, which are typically optimized for specialized data and tasks, to participate in higher-level reasoning and decision-making processes within agentic systems. Eywa can serve as a drop-in replacement for a single-agent pipeline (*EywaAgent*) or be integrated into existing multi-agent systems by replacing traditional agents with specialized agents (*EywaMAS*). We further investigate a planning-based orchestration framework in which a planner dynamically coordinates traditional agents and Eywa agents to solve complex tasks across heterogeneous data modalities (*EywaOrchestra*). We evaluate Eywa across a diverse set of scientific domains spanning physical, life, and social sciences. Experimental results demonstrate that Eywa improves performance on tasks involving structured and domain-specific data, while reducing reliance on language-based reasoning through effective collaboration with specialized foundation models.

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Figure 1: ***Eywa* extends current agentic systems. (Left)** Overall comparisons show that *EywaAgent*, *EywaMAS*, and *EywaOrchestra* achieve higher utility with lower token consumption than language-only baselines. **(Right)** Category-level results further show consistent gains in utility, token efficiency, and execution time across physical, life, and social science tasks.

Figure 1: Shows comprehensive experimental results demonstrating Eywa's superior utility-cost trade-off across physical, life, and social science domains with consistent improvements in utility, token efficiency, and execution time.

Limitations

- Requires expert-defined configurations for foundation model assignment in EywaAgent and EywaMAS (only EywaOrchestra provides automation)
- Limited to finite topology pool for tractability in dynamic orchestration, constraining system flexibility
- Depends on availability of domain-specific foundation models with structured interfaces, limiting applicability to domains without such models
- Evaluation restricted to two foundation models (Chronos for time series, TabPFN for tabular data), requiring validation on broader model types